

FGFR2 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR2 gene fusions from the msk_impact_50k_2026 study, covering 136 fusion events. 108/136 fusions (79.4%) were in-frame. 115/136 fusions (84.6%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.000424687$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR2.

1 Results summary

1.1 Fusion partner frequency

Fusion-partner frequency was tabulated across 136 analyzed fusion events, identifying 78 distinct fusion partner genes. The most frequent partner was BICC1, observed in 28 events.

1.2 Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 99/108 (91.7%) of in-frame fusions retained the domain, $p=0.000424687$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

1.3 Domain disruption

Domain-disruption analysis was skipped: FGFR2 has no `disruption_required_domains` configured; domain-disruption analysis was skipped.

1.4 Cutpoint detection

Cutpoint detection scanned 134 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 732 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -25 aa from the nearest configured domain boundary at 757 aa.

1.5 Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: FGFR2 has no `group_field` configured; confidence-stats analysis was skipped.

1.6 Composite evidence score

Composite evidence score reported: `domain_disruption` sub-score excluded: the supplied result had no applicable p-value (e.g. `disruption_required_domains` not configured for this gene).

1.7 Exon retention

Exon retention reported: No target exon was supplied in `params` or `GeneConfig`.

1.8 Joint partner

Joint partner reported: FGFR2 has no `gene_pair` configured; joint-partner analysis was skipped.

2 Tables

2.1 composite_score composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty	Components_applicable	Composite_score	Rank
BICC1	28	0.20588235294117646	0.20043213737826426		0.9354064039408867		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.3602624297400743	1
TACC2	8	0.058823529411764705	0.20043213737826426		0.9314655172413793		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.29639274393471765	2
ANKRD11	1	0.007352941176470588	0.20043213737826426		1.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28901988099643894	3
NRG3	2	0.014705882352941176	0.20043213737826426		0.9698275862068966		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2857056242592613	4
NOL4	4	0.029411764705882353	0.20043213737826426		0.9392241379310344		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2854502634942656	5
KIAA1217	4	0.029411764705882353	0.20043213737826426		0.9387931034482758		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28535789896224595	6
TBC1D4	3	0.022058823529411766	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28239136752208366	7
CTNNA3	3	0.022058823529411766	0.20043213737826426		0.9390804597701149		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28226821481272407	8
SHTN1	3	0.022058823529411766	0.20043213737826426		0.9385057471264368		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.28214506210336443	9
INA	2	0.014705882352941176	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	10
SORBS1	2	0.014705882352941176	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	11
TTC28	2	0.014705882352941176	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	12
WAC	2	0.014705882352941176	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.279240107017882	13
AHCYL1	2	0.014705882352941176	0.20043213737826426		0.9387931034482758		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2790553779538426	14
BAIAP2L1	2	0.014705882352941176	0.20043213737826426		0.9387931034482758		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2790553779538426	15
NRBF2	2	0.014705882352941176	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2788706488898032	16
OPTN	2	0.014705882352941176	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2788706488898032	17
ABLIM1	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	18
ADAM9	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	19
AMOT	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	20
ATAD1	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	21
ATP6V1D	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	22
BICD1	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	23
BNIP2	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CAT	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	25
CCDC6	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	26
CEACAM7	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	27
CEP112	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	28
DDX21	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	29
DIABLO	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	30
GAB2	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	31
HSPA12A	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	32
KLHL29	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	33
MARVELD3	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	34
MYO18B	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	35
NRAP	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	36
PDZRN4	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	37
PKP4	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	38
POC1B	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	39
PRDX3	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	40
RNF145	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	41
RPAP3	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	42
SH3GLB1	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	43
SHC2	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	44
SKI	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	45
SRRT	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	46
TFEC	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	47
VCL	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	48
VPS26A	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	49
YTHDF3	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	50
ZMYM4	1	0.007352941176470588	0.20043213737826426		0.9396551724137931		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27608884651368026	51
BTBD16	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	52

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
CCDC186	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	53
CCDC73	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	54
KIF14	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	55
L3MBTL3	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	56
LAMC1	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	57
MBIP	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	58
MYH15	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	59
MYH9	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	60
MYLK	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	61
NID2	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	62
PAH	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	63
PALLD	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	64
PPP1R21	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	65
RABGAP1L	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	66
ROCK1	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	67
SEPTIN10	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	68
STAT4	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	69
UBP1	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	70
ZFAT	1	0.007352941176470588	0.20043213737826426		0.9379310344827586		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27571938838560145	71
FOXP1	2	0.014705882352941176	0.20043213737826426		0.9103448275862069		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.27295931884054214	72
ADAM12	1	0.007352941176470588	0.20043213737826426		0.903448275862069		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2683302258240251	73
KCNH7	1	0.007352941176470588	0.20043213737826426		0.8862068965517241		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.26463564454323696	74
ATE1	3	0.022058823529411766	0.20043213737826426		0.8080459770114943		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.2541893970787339	75
PRKAG2	1	0.007352941176470588	0.20043213737826426				['domain_retention', 'recurrence']	0.09511621217728589	76
ST8SIA6	1	0.007352941176470588	0.20043213737826426				['domain_retention', 'recurrence']	0.09511621217728589	77
GRID1	1	0.007352941176470588	0.20043213737826426		0.0		['cutpoint_proximity', 'domain_retention', 'recurrence']	0.07473416671072464	78

2.2 cutpoint_detection cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
152	0	115	1	18	0.0	0.1417910447761194	0.8483511974119787
469	0	115	2	17	0.0	0.019189765458422176	1.7169303332757584
676	0	115	3	16	0.0	0.0024714091878270984	2.607055343103334
732	0	115	5	14	0.0	3.4829019675778254e-05	4.458058749354329
767	63	52	15	4	0.3230769230769231	0.07677359623872276	1.1147881156383952
768	111	4	19	0	0.0	1.0	-0.0
798	112	3	19	0	0.0	1.0	-0.0
801	113	2	19	0	0.0	1.0	-0.0
802	114	1	19	0	0.0	1.0	-0.0

2.3 domain_retention frame_domain_contingency_table

99	16
9	10

2.4 domain_retention domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained
kinase	Protein tyrosine and serine/threonine kinase	134	115	3	16	0.09006270188105643

2.5 frequency Partner_gene_counts

Partner_gene	Event_count
ABLIM1	1
ADAM12	1
ADAM9	1
AHCYL1	2
AMOT	1
ANKRD11	1
ATAD1	1
ATE1	3
ATP6V1D	1
BAIAP2L1	2
BICC1	28
BICD1	1
BNIP2	1
BTBD16	1
CAT	1
CCDC186	1
CCDC6	1
CCDC73	1
CEACAM7	1
CEP112	1

Partner_gene	Event_count
CTNNA3	3
DDX21	1
DIABLO	1
FOXP1	2
GAB2	1
GRID1	1
HSPA12A	1
INA	2
KCNH7	1
KIAA1217	4
KIF14	1
KLHL29	1
L3MBTL3	1
LAMC1	1
MARVELD3	1
MBIP	1
MYH15	1
MYH9	1
MYLK	1
MYO18B	1
NID2	1
NOL4	4
NRAP	1
NRBF2	2
NRG3	2
OPTN	2
PAH	1
PALLD	1
PDZRN4	1
PKP4	1
POC1B	1
PPP1R21	1
PRDX3	1
PRKAG2	1
RABGAP1L	1
RNF145	1
ROCK1	1
RPAP3	1
SEPTIN10	1
SH3GLB1	1
SHC2	1
SHTN1	3
SKI	1
SORBS1	2
SRRT	1
ST8SIA6	1
STAT4	1
TACC2	8
TBC1D4	3
TFEC	1
TTC28	2
UBP1	1
VCL	1
VPS26A	1
WAC	2
YTHDF3	1
ZFAT	1
ZMYM4	1

2.6 Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_insert	is_retained	is_in_domain	disrupted_domain	description	annotation	effect_on_fusion
EVT-P-0021826-T01-IM6-1	P-0021826-T01-IM6		ANKRD11-FGFR2	ANKRD11	unknown	five_prime	123243799	17	732	True	disrupted	0.909747292	16726				Protein tyrosine and serine/threonine kinase	FGFR2 (NM_000141) - ANKRD11 (NM_013275) rearrangement: t(10;16)(q26.13;q24.3)(chr10:g.123243799 chr16:g.16726)	NA	
EVT-P-0013241-T01-IM5-2	P-0013241-T01-IM5		ATP6V1D-FGFR2	ATP6V1D	in-frame	five_prime	123242048	17	767	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			Fusion {FGFR2:ANKRD11} FGFR2 (NM_000141) - ATP6V1D (NM_015994) rearrangement: t(10;14)(q26.13;q23.2)(chr10:g.123243799 chr14:g.14767)	NA	
EVT-P-0009512-T01-IM5-3	P-0009512-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123240449	18	768	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			in frame (FGFR2-ATP6V1D) Note: The FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal rearrangement event is an inversion which results in the fusion of FGFR2 exons 1-17 with BICC1 exons 3-21. The fusion is predicted to be in frame and includes the kinase domain of FGFR2. Protein fusion: in frame (FGFR2-BICC1)	NA	
EVT-P-0002675-T01-IM3-4	P-0002675-T01-IM3		BICC1-FGFR2	BICC1	in-frame	five_prime	123241096	18	768	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR2-BICC1)	NA	
EVT-P-0029067-T01-IM6-5	P-0029067-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123240197	18	768	True	retained	1.0	False		Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 3-21): c.2302-67:FGFR2_c.238-6757:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	main_retained	donor_insertion	retained_deletion	domain	disrupted_domain	sequence_annotation	effect_on_fusion	
EVT-P-0003177-T01-IM5-6	P-0003177-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123242188	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			NA Protein fusion: in frame (FGFR2-BICC1)	NA	
EVT-P-0025807-T01-IM6-7	P-0025807-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123242181	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 3-21): c.2301+895:FGFR2_c.238-35107:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	238-
EVT-P-0025250-T01-IM6-8	P-0025250-T01-IM6		BICC1-FGFR2	BICC1	in-frame	three_prime	123240491	18	768	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused to BICC1 exons 16-21): c.2302-956:FGFR2_c.2182-297:BICC1inv Protein Fusion: in frame {BICC1:FGFR2}	NA (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BIC	
EVT-P-0032033-T01-IM6-9	P-0032033-T01-IM6		BICC1-FGFR2	BICC1	in-frame	five_prime	123241685	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BIC	NA (NM_000141) - BICC1 (NM_001080512) Fusion (FGFR2 exon 18 fused with BICC1 exon3) : c.2301+1527:FGFR2_c.237+38877:BIC	
EVT-P-0005538-T01-IM5-10	P-0005538-T01-IM5		BICC1-FGFR2	BICC1	in-frame	five_prime	123242753	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase			FGFR2 (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:FGFR2_c.2182-276:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA (NM_000141) - BICC1 (NM_001080512) reciprocal fusion (FGFR2 exons 1-17 with BICC1 exons 16-21): c.2301+459:FGFR2_c.2182-276:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	2182-

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1_pos	protein_2_pos	breakpoint_4	main_ref	chr1	start	end	domain	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0037617-T01-IM6-16	P-0037617-T01-IM6		BICC1-FGFR2	BICC1	in-frame	three_prime	123241302	18	768	True	lost	0.0	False				Protein tyrosine and serine/threonine kinase		BICC1 (NM_001080512) - FGFR2 (NM_000141) rearrangement: c.238-34060:BICC1_c.2302-1767:FGFR2inv Protein Fusion: in frame {BICC1:FGFR2} FGFR2 NA (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26.13;p13)(chr10:g.123242534-1767:FGFR2inv Protein fusion: in frame (FGFR2-CAT) FGFR2 NA (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+1670:FGFR2_c.582+6426:CCDC6 Protein Fusion: in frame {FGFR2:CCDC6} FGFR2 NA (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.12324199-1767:FGFR2inv Protein Fusion: in frame {FGFR2:CEACAM7} t(17;10)(q24.1;p11.2)(chr17:g.64023428-1767:FGFR2inv Protein fusion: in frame (CCDC46-FGFR2) FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0015432-T01-IM6-17	P-0015432-T01-IM6		CAT-FGFR2	CAT	in-frame	five_prime	123242534	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) - CAT (NM_001752) rearrangement: t(10;11)(q26.13;p13)(chr10:g.123242534-1767:FGFR2inv Protein fusion: in frame (FGFR2-CAT) FGFR2 NA (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+1670:FGFR2_c.582+6426:CCDC6 Protein Fusion: in frame {FGFR2:CCDC6} FGFR2 NA (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.12324199-1767:FGFR2inv Protein Fusion: in frame {FGFR2:CEACAM7} t(17;10)(q24.1;p11.2)(chr17:g.64023428-1767:FGFR2inv Protein fusion: in frame (CCDC46-FGFR2) FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0027305-T01-IM6-18	P-0027305-T01-IM6		CCDC6-FGFR2	CCDC6	in-frame	five_prime	123241542	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) - CCDC6 (NM_005436) fusion (FGFR2 exons 1-17 fused to CCDC6 exons 4-9): c.2301+1670:FGFR2_c.582+6426:CCDC6 Protein Fusion: in frame {FGFR2:CCDC6} FGFR2 NA (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.12324199-1767:FGFR2inv Protein Fusion: in frame {FGFR2:CEACAM7} t(17;10)(q24.1;p11.2)(chr17:g.64023428-1767:FGFR2inv Protein fusion: in frame (CCDC46-FGFR2) FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0027741-T01-IM6-19	P-0027741-T01-IM6		CEACAM7-FGFR2	CEACAM7	unknown	five_prime	123241990	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) rearrangement: t(10;19)(q26.13;q13.2)(chr10:g.12324199-1767:FGFR2inv Protein Fusion: in frame {FGFR2:CEACAM7} t(17;10)(q24.1;p11.2)(chr17:g.64023428-1767:FGFR2inv Protein fusion: in frame (CCDC46-FGFR2) FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0003724-T01-IM5-20	P-0003724-T01-IM5		CEP112-FGFR2	CEP112	in-frame	three_prime	123241923	17	767	True	lost	0.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0019723-T01-IM6-22	P-0019723-T01-IM6		CTNNA3-FGFR2	CTNNA3	in-frame	five_prime	123243018	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		
EVT-P-0011516-T01-IM5-23	P-0011516-T01-IM5		CTNNA3-FGFR2	CTNNA3	in-frame	three_prime	123241940	17	767	True	lost	0.0	False				Protein tyrosine and serine/threonine kinase		FGFR2 NA (NM_000141) - CTNNA3 (NM_013266) rearrangement: c.2301+194:FGFR2_c.1885-25982:CTNNA3del Protein Fusion: in frame {FGFR2:CTNNA3} CTNNA3 NA (NM_013266) -FGFR2 (NM_000141) Rearrangement: c.1048-8479:CTNNA3_c.2301+1272:FGFR2dup Protein fusion: in frame (CTNNA3-FGFR2)		

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_pos	protein_pos	protein_pos	main_retained	other_fragments	disrupted_domains	disrupted_domains	disrupted_domains	disrupted_domains	disruption_annotation	disruption_annotation
EVT-P-0020410-T01-IM6-24	P-0020410-T01-IM6		DDX21-FGFR2	DDX21	in-frame	five_prime	123243048	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - DDX21 (NM_004728) fusion (FGFR2 exons 1-17 fused to DDX21 exons 2-15): c.2301+164 FGFR2_c.88-860:DDX21inv Protein Fusion: in frame {FGFR2:DDX21} FGFR2 NA (NM_000141) - ABLIM1 (NM_002313) fusion: c.2301+611.FGFR2_c.895-12244:ABLIM1del Protein Fusion: in frame {FGFR2:ABLIM1} FGFR2 NA (NM_000141) - ADAM12 (NM_003474) rearrangement: c.2026:FGFR2_c.1618+5260:ADAM12del Protein Fusion: mid-exon {FGFR2:ADAM12} FGFR2 NA (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10)(p11.22;q26.13)(chr8:g.38901161::: chr10:g.110531185::)	
EVT-P-0018869-T03-IM6-26	P-0018869-T03-IM6		FGFR2-ABLIM1	ABLIM1	in-frame	five_prime	123242601	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				{FGFR2:ABLIM1} FGFR2 NA (NM_000141) - ABLIM1 (NM_002313) fusion: c.2301+611.FGFR2_c.895-12244:ABLIM1del Protein Fusion: in frame {FGFR2:ABLIM1} FGFR2 NA (NM_000141) - ADAM12 (NM_003474) rearrangement: c.2026:FGFR2_c.1618+5260:ADAM12del Protein Fusion: mid-exon {FGFR2:ADAM12} FGFR2 NA (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10)(p11.22;q26.13)(chr8:g.38901161::: chr10:g.110531185::)	
EVT-P-0026275-T01-IM6-27	P-0026275-T01-IM6		FGFR2-ADAM12	ADAM12	unknown	five_prime	123246899	15	676	False	disrupted	0.7075812274368231		Protein tyrosine and serine/threonine kinase				{FGFR2:ADAM12} FGFR2 NA (NM_000141) - ADAM9 (NM_003816) fusion: t(8;10)(p11.22;q26.13)(chr8:g.38901161::: chr10:g.110531185::)	
EVT-P-0020164-T01-IM6-28	P-0020164-T01-IM6		FGFR2-ADAM9	ADAM9	in-frame	five_prime	123242738	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				{FGFR2:ADAM9} FGFR2 NA (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AH-CYL1 exons 2-17): t(1;10)(p13.3;q26.13)(chr1:g.110531185::: chr10:g.123241762::)	
EVT-P-0018920-T01-IM6-29	P-0018920-T01-IM6		FGFR2-AHCYL1	AHCYL1	in-frame	five_prime	123242706	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				{FGFR2:AHCYL1} FGFR2 NA (NM_000141) - AHCYL1 (NM_006621) fusion (FGFR2 exons 1-17 fused in-frame with AH-CYL1 exons 2-17): t(1;10)(p13.3;q26.13)(chr1:g.110531185::: chr10:g.123241762::)	
EVT-P-0000644-T01-IM3-30	P-0000644-T01-IM3		FGFR2-AHCYL1	AHCYL1	in-frame	three_prime	123239745	18	768	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase			{FGFR2:AHCYL1} FGFR2 NA (NM_000141) - AMOT (NM_001113490) Rearrangement : t(10;X)(q26.13.1;q23)(chr10:g.123241762::: chr10:g.123241762::)	
EVT-P-0012666-T01-IM5-31	P-0012666-T01-IM5		FGFR2-AMOT	AMOT	in-frame	five_prime	123241762	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				{FGFR2:AMOT} FGFR2 NA (NM_000141) - AMOT (NM_001113490) Rearrangement : t(10;X)(q26.13.1;q23)(chr10:g.123241762::: chr10:g.123241762::)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	is_main_ref	is_disrupted	description	annotation	effect_on_fra
EVT-P-0047504-T01-IM6-32	P-0047504-T01-IM6		FGFR2-ATAD1	ATAD1	in-frame	five_prime	123241680	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - ATAD1 (NM_032810) fusion: c.2301+1532:FGFR2_c.831+3993:ATAD1 Protein Fusion: in frame {FGFR2:ATAD1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908:FGFR2_c.1379-520:ATE1dup Protein Fusion: out of frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - BA-IAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::c	NA
EVT-P-0017424-T01-IM6-33	P-0017424-T01-IM6		FGFR2-ATE1	ATE1	out-of-frame	five_prime	123240443	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.2302-908:FGFR2_c.1379-520:ATE1dup Protein Fusion: out of frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - BA-IAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::c	NA
EVT-P-0063512-T01-IM7-35	P-0063512-T01-IM7		FGFR2-ATE1	ATE1	unknown	five_prime	123263336	10	469	False	lost	0.0	False	Protein tyrosine and serine/threonine kinase	Protein tyrosine and serine/threonine kinase	FGFR2 (NM_000141) - ATE1 (NM_001001976) rearrangement: c.1407:FGFR2_c.943-1893:ATE1dup Protein Fusion: mid-exon {FGFR2:ATE1} FGFR2 NA (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - BA-IAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::c	NA
EVT-P-0063625-T01-IM7-36	P-0063625-T01-IM7		FGFR2-ATE1	ATE1	in-frame	five_prime	123242847	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - ATE1 (NM_001001976) fusion: c.2301+365:FGFR2_c.975+6106:ATE1dup Protein Fusion: in frame {FGFR2:ATE1} FGFR2 NA (NM_000141) - BA-IAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::c	NA
EVT-P-0052794-T01-IM6-38	P-0052794-T01-IM6		FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123240813	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98001949::c	NA
EVT-P-0059296-T01-IM7-39	P-0059296-T01-IM7		FGFR2-BAIAP2L1	BAIAP2L1	in-frame	five_prime	123242027	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - BAIAP2L1 (NM_018842) fusion: t(7;10)(q22.1;q26.13)(chr7:g.98029133::c	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	cond_1	fract_1	main_ref	cond_2	fract_2	disrupted_domains	description	annotation	effect_on_fra
EVT-P-0006885-T01-IM5-48	P-0006885-T01-IM5		FGFR2-BICC1	BICC1	in-frame	five_prime	123240587	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) rearrangement: c.2302-1052:FGFR2_c.238-5239:BICC1inv Protein fusion: in frame (FGFR2-BICC1)	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0051773-T01-IM6-49	P-0051773-T01-IM6		FGFR2-BICC1	BICC1	in-frame	five_prime	123242828	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+384:FGFR2_c.237+13961:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0049133-T02-IM6-50	P-0049133-T02-IM6		FGFR2-BICC1	BICC1	in-frame	five_prime	123242098	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+1114:FGFR2_c.237+29879:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0053822-T01-IM6-51	P-0053822-T01-IM6		FGFR2-BICC1	BICC1	in-frame	five_prime	123243033	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2301+179:FGFR2_c.238-1932:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0021327-T01-IM6-52	P-0021327-T01-IM6		FGFR2-BICC1	BICC1	in-frame	five_prime	123240344	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion (FGFR2 exons 1-17 fused with BICC1 exons 16-21): c.2302-1613:FGFR2_c.237+32254:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0060245-T01-IM7-53	P-0060245-T01-IM7		FGFR2-BICC1	BICC1	unknown	five_prime	123239377	18	820	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2460:FGFR2_c.1179+1103:BICC1inv Protein Fusion: mid-exon {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	
EVT-P-0068033-T01-IM7-54	P-0068033-T01-IM7		FGFR2-BICC1	BICC1	in-frame	five_prime	123239673	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase						FGFR2 (NM_000141) - BICC1 (NM_001080512) fusion: c.2302-138:FGFR2_c.238-40422:BICC1inv Protein Fusion: in frame {FGFR2:BICC1}	NA (NM_000141) - BICC1 (NM_001080512)	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	inframe_ref	inframe_host	inframe_donor	inframe_domain	disrupted_domain	description_annotation	clinical_effect_on_fra
EVT-P-0064855-T01-IM7-115	P-0064855-T01-IM7		FGFR2-MBIP	MBIP	in-frame	five_prime	123240070	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - MBIP (NM_016586) fusion: t(10;14)(q26.13;q13.3)(chr10:g.123240070 chr14:g.100000000) Protein Fusion: in frame {FGFR2:MBIP} FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.13;q26.13)(chr3:g.108134575 chr10:g.123240070) Protein Fusion: in frame {FGFR2:MYH15} FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612 chr10:g.123240070) Transcript Fusion {FGFR2:MYLK} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0030844-T01-IM6-116	P-0030844-T01-IM6		FGFR2-MYH15	MYH15	in-frame	five_prime	123240665	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - MYH15 (NM_014981) rearrangement: t(3;10)(q13.13;q26.13)(chr3:g.108134575 chr10:g.123240665) Protein Fusion: in frame {FGFR2:MYH15} FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612 chr10:g.123240665) Transcript Fusion {FGFR2:MYLK} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0027623-T01-IM6-117	P-0027623-T01-IM6		FGFR2-MYLK	MYLK	unknown	five_prime	123240959	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) rearrangement: t(3;10)(q21.1;q26.13)(chr3:g.123525612 chr10:g.123240959) Transcript Fusion {FGFR2:MYLK} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0038235-T01-IM6-118	P-0038235-T01-IM6		FGFR2-NOL4	NOL4	in-frame	five_prime	123240418	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - NOL4 (NM_003787) fusion: t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123240418 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0012423-T02-IM6-119	P-0012423-T02-IM6		FGFR2-NOL4	NOL4	in-frame	five_prime	123242055	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - NOL4 (NM_003787) fusion (FGFR2 exons 1-17 fused in-frame with NOL4 exons 7-11): t(10;18)(q26.13;q12.1)(chr10:g.123242055 chr18:g.100000000) Protein Fusion: in frame {FGFR2:NOL4} FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0006328-T02-IM6-120	P-0006328-T02-IM6		FGFR2-NRBF2	NRBF2	in-frame	five_prime	123241246	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - NRBF2 (NM_030759) fusion: c.2302-1711:FGFR2_c.157-363NRBF2inv Protein Fusion: in frame {FGFR2:NRBF2} FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	
EVT-P-0058339-T01-IM6-121	P-0058339-T01-IM6		FGFR2-PAH	PAH	in-frame	five_prime	123240026	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - PAH (NM_000277) fusion: t(10;12)(q26.13;q23.2)(chr10:g.123240026 chr12:g.100000000) Protein Fusion: in frame {FGFR2:PAH}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr1_start	chr1_end	chr2_start	chr2_end	disrupted_domain	description	annotation	effect_on_fra
EVT-P-0015500-T01-IM6-122	P-0015500-T01-IM6		FGFR2-PALLD	PALLD	in-frame	five_prime	123239869	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - PALLD (NM_001166108) Rearrangement: t(4;10)(q33;q26.13)(chr4:g.169605121::chr10:g.110325395::cl	NA	
EVT-P-0021448-T03-IM6-123	P-0021448-T03-IM6		FGFR2-PKP4	PKP4	in-frame	five_prime	123242661	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - PKP4 (NM_003628) rearrangement: t(2;10)(q24.1;q26.13)(chr2:g.159395413::chr10:g.110325395::cl	NA	
EVT-P-0059669-T01-IM7-125	P-0059669-T01-IM7		FGFR2-POC1B	POC1B	in-frame	five_prime	123242798	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - POC1B (NM_001199777.2) fusion: t(10;12)(q26.13;q21.33)(chr10:g.123242798::chr12:g.110325395::cl	NA	
EVT-P-0008657-T01-IM5-126	P-0008657-T01-IM5		FGFR2-PPP1R21	PPP1R21	in-frame	five_prime	123240684	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - PPP1R21 (NM_152994) Rearrangement: t(2;10)(2p16.3;10q26.13)(chr2:g.4872186::chr10:g.110325395::cl	NA	
EVT-P-0057247-T02-IM7-127	P-0057247-T02-IM7		FGFR2-PRDX3	PRDX3	in-frame	five_prime	123242894	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - PRDX3 (NM_006793) fusion: c.2301+318;FGFR2_c.36+287:PRDX3d	NA	
EVT-P-0011394-T01-IM5-129	P-0011394-T01-IM5		FGFR2-RABGAP1L	RABGAP1L	in-frame	five_prime	123240065	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - RABGAP1L (NM_014857) rearrangement: t(1;10)(q25.1;q26.13)(chr1:g.174863117::chr10:g.110325395::cl	NA	
EVT-P-0012639-T01-IM5-130	P-0012639-T01-IM5		FGFR2-SEPTIN10	SEPTIN10	unknown	five_prime	123240798	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase					FGFR2 (NM_000141) - SEPT10 (NM_144710) rearrangement: t(2;10)(q13;q26.13)(chr2:g.110325395::chr10:g.110325395::cl	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint	breakpoint_pos	breakpoint_refseq	protein_pos	breakpoint_pos_refseq	domain_retained	domain_lost	domain_dropped	disrupted_domains	annotation	effect_on_function
EVT-P-0057746-T01-IM6-149	P-0057746-T01-IM6		FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128-402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2} FGFR2 NA (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128-402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	
EVT-P-0057746-T02-IM6-150	P-0057746-T02-IM6		FGFR2-TACC2	TACC2	in-frame	five_prime	123241998	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - TACC2 (NM_206862) fusion: c.2301+1214:FGFR2_c.8128-402:TACC2inv Protein Fusion: in frame {FGFR2:TACC2}	
EVT-P-0022337-T06-IM6-151	P-0022337-T06-IM6		FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723-17241724:17241724-17241724) Protein Fusion: in frame {FGFR2:TBC1D4}	
EVT-P-0022337-T04-IM6-152	P-0022337-T04-IM6		FGFR2-TBC1D4	TBC1D4	in-frame	five_prime	123241723	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - TBC1D4 (NM_014832) fusion: t(10;13)(q26.13;q22.2)(chr10:g.123241723-17241724:17241724-17241724) Protein Fusion: in frame {FGFR2:TBC1D4}	
EVT-P-0005291-T01-IM5-153	P-0005291-T01-IM5		FGFR2-TFEC	TFEC	unknown	five_prime	123241729	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_022970) - TFEC (NM_012252) Rearrangement: t(10;7)(q26.13;q31.2)(chr10:g.123241729-17241729:17241729-17241729) Transcript fusion (FGFR2-TFEC)	
EVT-P-0016640-T01-IM6-154	P-0016640-T01-IM6		FGFR2-TTC28	TTC28	in-frame	five_prime	123241944	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000142) - TTC28 (NM_001145418) Rearrangement: t(10;22)(q26.11;q12.1)(chr10:g.123241944-19441944:19441944-19441944) Protein Fusion: in frame {FGFR2:TTC28}	
EVT-P-0051558-T01-IM6-155	P-0051558-T01-IM6		FGFR2-TTC28	TTC28	in-frame	five_prime	123243158	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase		FGFR2 (NM_000141) - TTC28 (NM_001145418) fusion: t(10;22)(q26.13;q12.1)(chr10:g.123243158-158158:158158-158158) Protein Fusion: in frame {FGFR2:TTC28}	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_pos	protein_pos_2	protein_pos_3	main_retention	domain_pos	domain_pos_2	domain_pos_3	disrupted_domains	domains_annotation	mutation_site	effect_on_fra
EVT-P-0016195-T01-IM6-164	P-0016195-T01-IM6		GAB2-FGFR2	GAB2	in-frame	five_prime	123242812	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - GAB2 (NM_080491) fusion (FGFR2 exons 1-17 fused with GAB2 exons 2-10) : t(10;11)(q26.11;q14.1)(chr10:g.123242812) Protein Fusion: in frame {FGFR2:GAB2}	NA	
EVT-P-0013756-T01-IM5-165	P-0013756-T01-IM5		INA-FGFR2	INA	in-frame	three_prime	123242695	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase			INA (NM_032727) - FGFR2 (NM_000141) Rearrangement : c.1066-3500:INA_c.2301+517:FGFR2inv Protein fusion: in frame (INA-FGFR2)	NA	
EVT-P-0013756-T03-IM6-166	P-0013756-T03-IM6		INA-FGFR2	INA	in-frame	three_prime	123242695	17	767	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase			INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500:INA_c.2301+517:FGFR2inv Protein Fusion: in frame {INA:FGFR2}	NA	
EVT-P-0002810-T01-IM3-169	P-0002810-T01-IM3		KIAA1217-FGFR2	KIAA1217	in-frame	five_prime	123240874	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				INA (NM_032727) - FGFR2 (NM_000141) rearrangement: c.1066-3500:INA_c.2301+517:FGFR2inv Protein Fusion: in frame {INA:FGFR2}	NA	
EVT-P-0000386-T02-IM5-170	P-0000386-T02-IM5		MARVELD3-FGFR2	MARVELD3	unknown	five_prime	123243101	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				MARVELD3 (NM_001017967) - FGFR2 (NM_000141) rearrangement: t(16;10)(16q22.2;10q26.13)(chr16:g.71660650::chr10:123243101) Antisense fusion FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240412) Protein Fusion: in frame {FGFR2:MYH9}	NA	
EVT-P-0024094-T01-IM6-172	P-0024094-T01-IM6		MYH9-FGFR2	MYH9	in-frame	five_prime	123240418	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - MYH9 (NM_002473) rearrangement: t(10;22)(q26.13;q12.3)(chr10:g.123240412) Protein Fusion: in frame {FGFR2:MYH9}	NA	
EVT-P-0032208-T01-IM6-173	P-0032208-T01-IM6		MYO18B-FGFR2	MYO18B	in-frame	five_prime	123241929	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 (NM_000141) - MYO18B (NM_032608) rearrangement: t(10;22)(q26.13;q12.2)(chr10:g.123241929) Protein Fusion: in frame {FGFR2:MYO18B}	NA	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	donor_fragment	inframe	disrupted_domain	disrupted_domain	description_annotation	effect_on_fra
EVT-P-0022337-T01-IM6-181	P-0022337-T01-IM6		NRG3-FGFR2	NRG3	out-of-frame	three_prime	123241723	17	767	True	lost	0.0	False			Protein tyrosine and serine/threonine kinase		NRG3 (NM_001010848) - FGFR2 (NM_000141) Rearrangement: c.953+89877:NRG3_c.2301+1489:FGFR2 Protein Fusion: out of frame {NRG3:FGFR2} FGFR2 NA (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN} FGFR2 NA (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN} FGFR2 NA (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::c	
EVT-P-0032171-T03-IM6-182	P-0032171-T03-IM6		OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 NA (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN} FGFR2 NA (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN} FGFR2 NA (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::c	
EVT-P-0032171-T01-IM6-183	P-0032171-T01-IM6		OPTN-FGFR2	OPTN	in-frame	five_prime	123239789	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 NA (NM_000141) - OPTN (NM_021980) rearrangement: c.2302-254:FGFR2_c.553-1543:OPTNinv Protein Fusion: in frame {FGFR2:OPTN} FGFR2 NA (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::c	
EVT-P-0022781-T01-IM6-184	P-0022781-T01-IM6		PDZRN4-FGFR2	PDZRN4	in-frame	five_prime	123241997	17	767	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 NA (NM_000141) - PDZRN4 (NM_001164595) rearrangement: t(6;10)(q23.1;q26.1)(chr6:g.130408449::c	
EVT-P-0044768-T01-IM6-187	P-0044768-T01-IM6		RNF145-FGFR2	RNF145	out-of-frame	three_prime	123243127	17	767	True	lost	0.0	False			Protein tyrosine and serine/threonine kinase		RNF145 NA (NM_144726) - FGFR2 (NM_000141) rearrangement: t(5;10)(q33.3;q26.13)(chr5:g.158598146::c	
EVT-P-0006235-T01-IM5-188	P-0006235-T01-IM5		ROCK1-FGFR2	ROCK1	in-frame	five_prime	123240133	18	768	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase				FGFR2 NA (NM_000141) - ROCK1 (NM_005406) rearrangement: t(10;18)(q26.13;q11.1)(chr10:g.123240133::c	

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_1	breakpoint_2	breakpoint_3	protein_1	protein_2	breakpoint_4	main_ref	chr10	fract	variant	disrupted_domain	disrupted_desc	annotation	effect_on_fra
EVT-P-0039370-T01-IM6-189	P-0039370-T01-IM6		RPAP3-FGFR2	RPAP3	in-frame	three_prime	123242832	17	767	True	lost	0.0	False				Protein tyrosine and serine/threonine kinase		RPAP3 (NM_024604) - FGFR2 (NM_000141) rearrangement: t(10;12)(q26.13;q13.11)(chr10:g.123242832-767;chr12:g.100000000-100000000)	
EVT-P-0025074-T01-IM6-190	P-0025074-T01-IM6		SHTN1-FGFR2	SHTN1	in-frame	five_prime	123240122	18	768	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		Fusion: in frame {RPAP3:FGFR2} (NM_000141) - KIAA1598 (NM_001127211) Fusion (FGFR2 exon 17 with KIAA1598 exon 5): c.2302-587:FGFR2_c.267+2596:KIAA1598	
EVT-P-0032507-T01-IM6-191	P-0032507-T01-IM6		SORBS1-FGFR2	SORBS1	unknown	five_prime	123242268	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		Fusion: in frame {FGFR2:KIAA1598} (NM_000141) - FGFR2 (NM_000141) Rearrangement: c.2301+944:FGFR2_chr10:g.97250518del	
EVT-P-0022337-T01-IM6-192	P-0022337-T01-IM6		TBC1D4-FGFR2	TBC1D4	in-frame	five_prime	123241723	17	767	True	retained	1.0	False				Protein tyrosine and serine/threonine kinase		Transcript Fusion {FGFR2:SORBS1} (NM_000141) - TBC1D4 (NM_014832) Rearrangement: t(10;13)(q26.13;q22.2)(chr10:g.123241723-767;chr13:g.100000000-100000000)	
EVT-P-0021788-T01-IM6-193	P-0021788-T01-IM6		VPS26A-FGFR2	VPS26A	in-frame	three_prime	123242226	17	767	True	lost	0.0	False				Protein tyrosine and serine/threonine kinase		Fusion: in frame {FGFR2:TBC1D4} (NM_004896) - VPS26A (NM_000141) Rearrangement: c.4-1035:VPS26A_c.2301+986:FGFR2inv	

3 Figures

FGFR2 domain-retention track

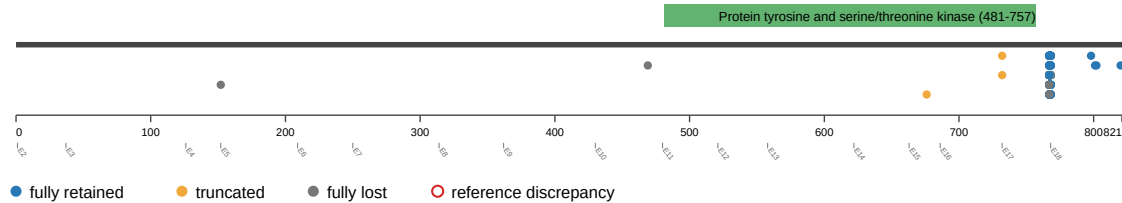
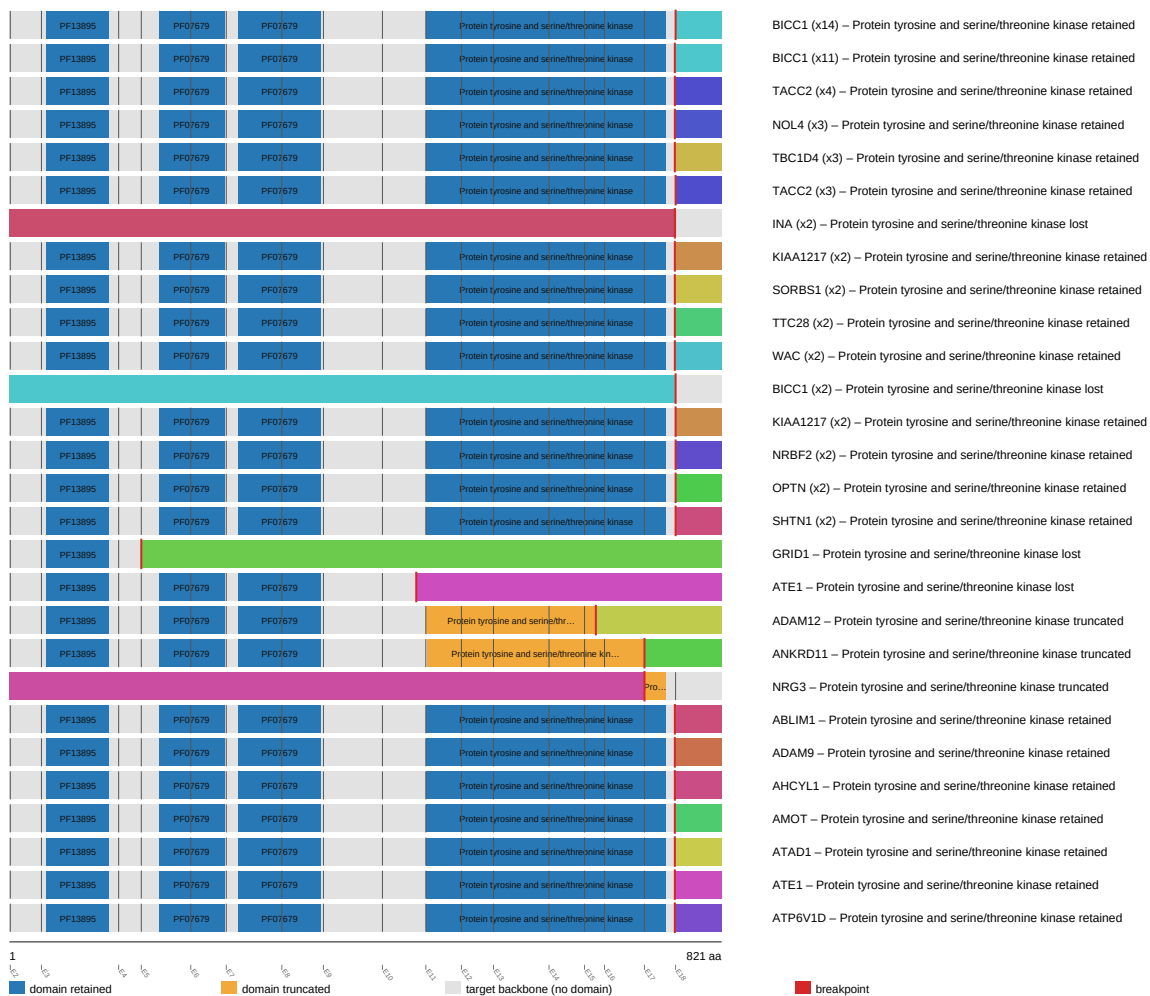


Figure 1: domain retention outliers

FGFR2 fusion-transcript schematic

partner block → breakpoint → retained FGFR2 portion (domain-colored, exon ticks)



Showing the top 28 of 92 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

Figure 2: fusion schematic

Reference vs msk_impact_50k_2026



Figure 3: reference comparison